

Cost-Effective Coordination for Peak Demand Reduction in Saudi Residential Buildings Using Physics-Informed Deep Reinforcement Learning

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Abstract

When multiple On/Off split air-conditioning units in Saudi residential buildings activate simultaneously, the resulting peak demand spike stresses the electrical grid and inflates monthly bills under the kingdom’s two-tier tariff (0.18 SAR/kWh $\leq$ 6,000 kWh; 0.30 SAR/kWh above). This paper proposes a Physics-Informed Proximal Policy Optimization (PI-PPO) framework that learns a *stationary* scheduling policy—applicable over an infinite time horizon without re-solving any optimization—to coordinate 18,500 BTU On/Off split units (1.8 kW input, EER 10.25) across multiple zones. Each zone is abstracted as a scheduling task with formally analyzed minimum utilization and feasibility conditions. The model incorporates inter-zone thermal coupling, enabling the scheduler to exploit thermal buffering through shared walls. PI-PPO embeds heat balance equations directly into the reinforcement learning reward, yielding a controller that maintains near-zero comfort violations (below 0.02°C in all tested scenarios)—a property absent from standard deep reinforcement learning methods. We further show that extending the comfort range by $\pm 1^\circ\text{C}$ (from 23–25°C to 22–26°C) reduces each zone’s minimum utilization by 36.9%. Simulations using EnergyPlus with Jeddah weather data across four months (January, April, July, October) show that PI-PPO reduces peak demand by 40–60% in cooling-dominant months (April–October), reaching 80% in mild January, and reduces July cost by 22.5% for a 5-zone villa, rising to 47.0% for a 20-zone compound with comfort extension. Ablation studies attribute 6.0 percentage points to physics-informed shaping, 4.5 pp to tiered-tariff awareness, 2.0 pp to inter-zone coupling, and 14.0 pp to comfort extension.

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Keywords: peak demand reduction; deep reinforcement learning; proximal policy optimization; demand-side management; HVAC scheduling; Saudi electricity tariff; thermal comfort

Submitted to: Journal of King Saud University – Engineering Sciences

1 Introduction

The building sector accounts for approximately 40% of global primary energy consumption [1]. In Saudi Arabia, air conditioning consumes over 70% of household electricity during summer [2], and the On/Off split unit—representing 60% of the residential market [2]—draws exactly 1.8 kW when its compressor runs or zero when it stops. When multiple such units activate simultaneously, the resulting peak demand spike has two consequences: it stresses the distribution grid, and it pushes cumulative monthly consumption past the 6,000 kWh threshold of the kingdom’s two-tier tariff [3], triggering a 67% marginal price increase from 0.18 to 0.30 SAR/kWh. The primary objective of this paper is to **reduce peak aggregate demand** through coordinated scheduling; the reduction in electricity cost follows as a direct economic consequence.

Model predictive control (MPC) has been applied to building cooling [4], [5], but solves a *finite-horizon* optimization at each control step—an approach that scales poorly with zone count due to combinatorial explosion [6]. Standard deep reinforcement learning (DRL) methods have achieved 5–21% energy savings in HVAC applications [7], [8], [9], but most optimize for flat-rate tariffs, lack physical consistency guarantees, and do not coordinate across zones. The physics-informed neural network (PINN) framework [10] has been adapted for building energy, achieving 78–90% *instantaneous peak power* reductions [11], [12], but under demand-charge tariffs (a per-kW penalty) rather than tiered per-kWh pricing.

This paper bridges these gaps with a Physics-Informed Proximal Policy Optimization (PI-PPO) framework that: (i) learns a *stationary* policy applicable over an infinite time horizon, eliminating the need to re-solve optimization at each step; (ii) embeds thermal dynamics into the reward to achieve near-zero comfort violations; (iii) incorporates inter-zone thermal coupling to exploit buffering through shared walls; and (iv) manages cumulative consumption relative to the tiered tariff threshold.

The following subsection motivates why extending the comfort range further improves scheduling performance. Section 2 reviews related work. Section 3 presents the system model with inter-zone coupling. Section 4 defines the task model and feasibility analysis. Section 5 describes the scheduling controller baseline. Section 6 details the PI-PPO methodology. Section 7 specifies the experimental setup with a worked numerical example. Section 8 presents seasonal results with ablation and scalability analyses. Section 9 concludes.

1.1 Why Extending the Comfort Range Reduces Peak Demand

Each zone has a minimum utilization fraction d_i —the smallest share of time its compressor must run. Widening the comfort band from $[23, 25]$ to $[22, 26]^\circ\text{C}$ reduces d_i by 36.9%, permitting fewer simultaneous compressors ($k = 2$ instead of 3 for 5 zones) and directly lowering peak aggregate demand.

A fixed comfort band was selected for two reasons. First, it enables the formal feasibility analysis (Section 4), which requires static bounds to derive the closed-form utilization d_i . Second, the chosen ranges align with ASHRAE Standard 55 [13], which supports adaptive comfort models where occupants accept wider bands in naturally ventilated or mixed-mode buildings. In the Saudi residential context, where indoor environments are mechanically cooled year-round, the adaptive comfort standard (ASHRAE 55 Section 5.4) permits operative temperatures up to 28°C at the mean outdoor temperatures observed in Jeddah. The extended range $[22, 26]^\circ\text{C}$ is therefore conservative relative to ASHRAE’s adaptive limits. Field validation of occupant acceptance in Saudi households remains necessary (see Limitations) and is identified as future work. Widening the thermostat deadband by $\pm 1^\circ\text{C}$ has been shown to reduce annual energy costs across all studied climates [14]. Section 4 formalizes this relationship.

2 Related Work

Saudi building energy. Saudi buildings consume approximately 80% of generated electricity, with cooling dominant [2]. The 2018 tariff reform saved USD 1.4 billion in fuel costs [15]. Peak demand-based retrofit optimization found efficiency measures cost-effective at 779 USD/kW [3]. Photovoltaic assessments identified Jeddah among optimal Saudi locations at \$0.047–\$0.057/kWh [16], with a 0.07 SAR/kWh feed-in tariff [17]. A mosque study demonstrated 30% reductions using EnergyPlus [18]. The minimum split-unit EER has risen from 7.5 to 11.8 since 2007 [2].

DRL for HVAC. A 2024 review found PPO and SAC best balanced stability and performance for HVAC control [19]. PPO achieved 5.27% energy savings over PID in a real data center [9], while DRL co-optimization of energy, comfort, and air quality reached 21.4% [8]. TD3 achieved 17% in multi-zone scenarios [7]. Hierarchical DRL optimized year-round operation [20]. RL validated for residential HVAC reported 8.8–26.3% savings [21]. Online RL with imitation learning proved reliable [6]. Transfer learning adapted policies across buildings [22], and multi-zone setpoint reset via DRL yielded significant reductions [23]. However, all these methods learn *episode-specific* policies under flat-rate tariffs without coordinating multiple zones against a tiered threshold.

Physics-informed learning and scheduling. Physics-informed neural networks combined with RC thermal models achieved 78% peak power reductions under demand-charge tariffs [11]. Modularized architectures with physical priors reached 90% [12]. Reviews confirmed growing PIML adoption for building energy [24], [25]. Neural-assisted scheduling

accelerated optimization [26]. Multi-agent hierarchical RL addressed distributed management [27] and peak shaving [28]. Neural architectures combined forecasting and scheduling in a single network [29]. Fast ML for building management achieved $36\times$ speedup [30]. Data-driven predictive control was reviewed as an alternative [31]. A systematic review confirmed PPO among the most effective algorithms [32]. The present work differs by targeting *tiered* tariffs with inter-zone coupling and an infinite-horizon stationary policy that maintains minimal comfort violations.

3 System Model

3.1 Thermal Dynamics with Inter-Zone Coupling

Consider a villa with n air-conditioned zones. The heat balance for zone i is:

$$C_i \frac{dx_i}{dt} = \underbrace{K_i(T_a - x_i)}_{\text{outdoor gain}} + \underbrace{\sum_{j \in \mathcal{N}(i)} K_{ij}(x_j - x_i)}_{\text{inter-zone transfer}} + \underbrace{Q_{\text{int},i}}_{\text{internal gains}} + \underbrace{Q_i(t)}_{\text{cooling}} \quad (1)$$

where C_i (kJ/K) is thermal capacity, K_i (kW/K) is conductance to outdoors, K_{ij} (kW/K) is conductance through the shared wall between adjacent zones i and j , T_a ($^{\circ}\text{C}$) is outdoor temperature, $Q_{\text{int},i}$ (kW) is internal gains, and $Q_i = -5.3$ kW when ON or 0 when OFF. The value 5.3 kW is the *effective sensible* cooling capacity: the nominal 18,500 BTU/hr rating (5.42 kW total) is derated by approximately 2% to account for indoor fan heat gain and the latent (dehumidification) fraction of the coil load under Jeddah’s humid coastal conditions, so that Q_i represents the heat actually removed from the sensible zone balance in Equation (1). The electrical draw is $P_i = 1.8$ kW (ON) or 0 (OFF). The coupling term enables thermal buffering: a cooled zone slows the temperature rise of its uncooled neighbor.

Neglecting coupling for analytical tractability (it is retained in the simulation), the single-zone dynamics become:

$$\frac{dx_i}{dt} = -a_i x_i + b_i, \quad a_i = \frac{K_i}{C_i}, \quad b_i = \frac{K_i T_a + Q_{\text{int},i} + Q_i}{C_i} \quad (2)$$

3.2 Jeddah Ambient Temperature Profiles

Figure 1 shows 24-hour profiles for the four evaluation months. The diurnal swing drives the coordination challenge: during the July afternoon peak ($\sim 43^{\circ}\text{C}$), all zones simultaneously demand maximum cooling, while January nights ($\sim 18^{\circ}\text{C}$) require no cooling at all.

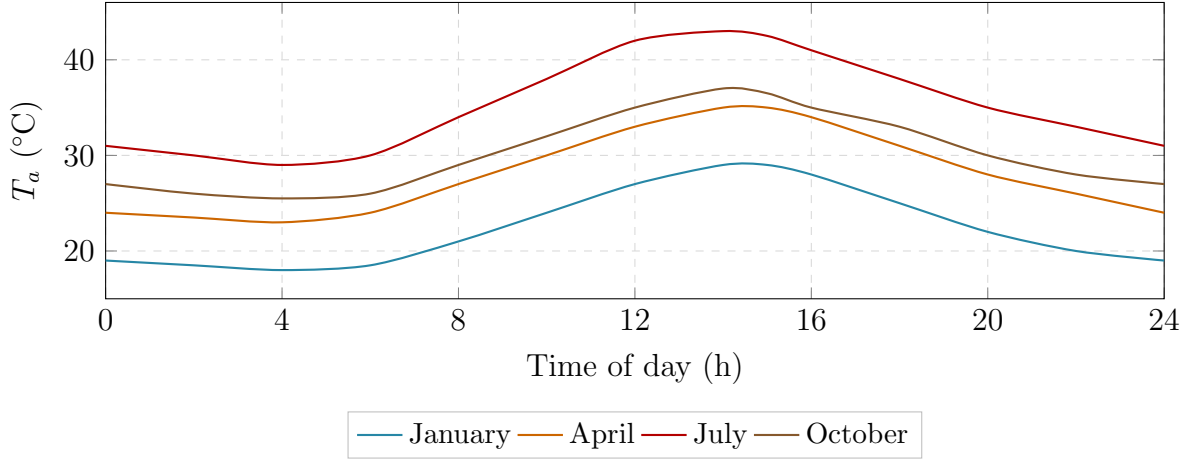


Figure 1: Jeddah 24-hour ambient temperature profiles (TMY3 data). July’s 43°C peak drives the most severe scheduling challenge.

3.3 Saudi Two-Tier Tariff

The residential tariff, unchanged since January 2018 [15], is:

$$\text{Bill} = \begin{cases} 0.18 \cdot E_{\text{tot}} & \text{if } E_{\text{tot}} \leq 6,000 \text{ kWh} \\ 0.18 \cdot 6,000 + 0.30 \cdot (E_{\text{tot}} - 6,000) & \text{otherwise} \end{cases} \quad (3)$$

The primary goal is peak demand reduction; this tariff quantifies the economic consequence. This tariff applies specifically to the *residential* customer classification as published by the Saudi Electricity and Cogeneration Regulatory Authority (ECRA). The model excludes value-added tax (VAT, currently 15%) and meter fixed charges, which are independent of scheduling decisions and do not affect relative savings comparisons. Future tariff reforms (e.g., time-of-use pricing under Vision 2030) would require extending the reward function but not the PI-PPO framework itself. The scheduling problem is stated as follows.

Peak Demand Minimization Problem. Given n zones with thermal dynamics (1), comfort bounds $[l_i, h_i]$, and tariff (3), compute a scheduling policy $\pi : \mathbf{s}(t) \rightarrow \mathbf{m}(t) \in \{0, 1\}^n$ that minimizes the peak aggregate electrical demand $P_{\text{peak}} = \max_t \sum_i 1.8 \cdot m_i(t)$ kW subject to $x_i(t) \in [l_i, h_i]$ for all i and all t . Sections 4–5 decompose this problem into feasibility analysis and scheduling; Section 6 solves it via PI-PPO.

4 Task Model and Feasibility Analysis

The feasibility analysis serves two roles in the PI-PPO framework. First, it provides the concurrency limit k that constrains the agent’s action space via the top- k projection, ensuring the scheduled number of active compressors never exceeds what is thermally feasible. Without this bound, the agent would need to discover the constraint empirically through violations. Second, the analysis formally quantifies the relationship between comfort band

width and scheduling capacity (Equation (7)), providing the theoretical justification for the comfort extension study in Section 8.

4.1 Task Abstraction

Following the green scheduling framework [33], [34], each zone i is abstracted as a scheduling task $\mathcal{T}_i = (x_i, l_i, h_i, X_{0,i}, \{\text{ON}, \text{OFF}\}, \text{dyn}_i)$, where x_i is temperature, $[l_i, h_i]$ is the comfort band, $X_{0,i} \subseteq [l_i, h_i]$ is the initial state set, and the dynamics follow:

$$\dot{x}_i = \begin{cases} -a_i^+ x_i + b_i^+ & \text{ON (compressor running, } Q_i = -5.3 \text{ kW)} \\ -a_i^- x_i + b_i^- & \text{OFF (compressor stopped, } Q_i = 0) \end{cases} \quad (4)$$

This is a first-order linear ODE. Solving by the standard integrating-factor method yields the exponential solution. If the compressor is OFF during $[t_1, t_2]$, the temperature rises as:

$$x_i(t) = \frac{b_i^-}{a_i^-} + \left(x_i(t_1) - \frac{b_i^-}{a_i^-} \right) e^{-a_i^-(t-t_1)} \quad (5)$$

The temperature asymptotically approaches $b_i^-/a_i^- \approx 36.2^\circ\text{C}$. If ON, it falls as:

$$x_i(t) = \frac{b_i^+}{a_i^+} + \left(x_i(t_1) - \frac{b_i^+}{a_i^+} \right) e^{-a_i^+(t-t_1)} \quad (6)$$

toward $b_i^+/a_i^+ \approx 15.0^\circ\text{C}$. The comfort thresholds must satisfy $b_i^+/a_i^+ < l_i < h_i < b_i^-/a_i^-$.

Figure 2 illustrates these dynamics for *both* comfort ranges. The strict-range trajectory (solid) oscillates between 23°C and 25°C with short, frequent cycles. The extended-range trajectory (dashed) oscillates between 22°C and 26°C with visibly longer OFF periods—directly demonstrating why wider bands reduce d_i .

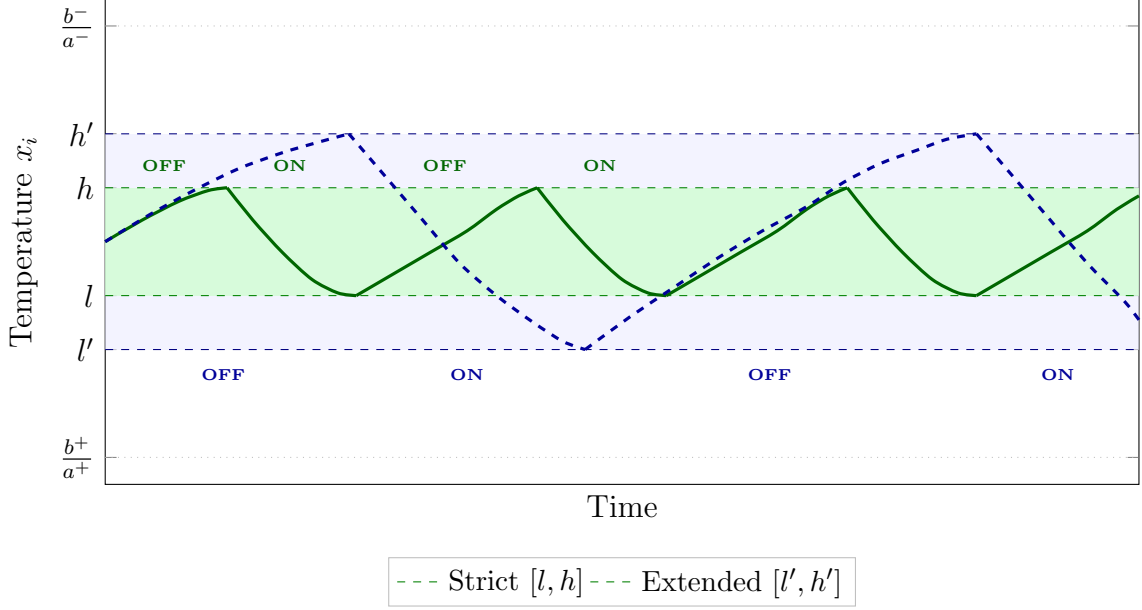


Figure 2: Cooling task dynamics (adapted from [33]). Green: strict $[l, h]$ range (~ 33 min OFF). Blue (dashed): extended $[l', h']$ range (~ 66 min OFF), reducing d_i from 0.480 to 0.303.

4.2 Minimum Utilization and Feasibility

The minimum utilization of zone i is:

$$d_i = \frac{a_i^- h_i - b_i^-}{(a_i^- h_i - b_i^-) - (a_i^+ h_i - b_i^+)} \quad (7)$$

where the numerator is the temperature rise rate at h_i when OFF and the denominator is the net rate difference. Note that d_i is evaluated at the *upper* threshold h_i because, in cooling mode, the critical event is the temperature reaching h_i (zone overheating), which forces immediate compressor activation. This is the cooling-mode counterpart of the heating-mode utilization evaluated at l_i in the original green scheduling formulation [33], where the critical event is the temperature dropping to the lower threshold. For n tasks with concurrency limit k :

Infeasibility condition. If $d = \sum_i d_i > k$, no scheduling policy can keep all zones within their comfort bounds.

Feasibility condition. If $d < k$ and at most k tasks start at h_i , a safe schedule exists. Since $\partial d_i / \partial h_i < 0$, extending from $h = 25$ to $h' = 26$ reduces d_i (Equation (7)) from 0.480 to 0.303 (a 36.9% reduction), as visible in Figure 2: wider bands permit longer OFF segments.

5 Scheduling Controller

The controller coordinates compressor modes $\mathbf{m} = [m_1, \dots, m_n]^\top \in \{0, 1\}^n$ subject to $\|\mathbf{m}\|_1 \leq k$ (resource constraint) and $\mathbf{x} \in [l_1, h_1] \times \dots \times [l_n, h_n]$ (safety). The coupled

system dynamics is:

$$\dot{\mathbf{x}} = \mathbf{A}_0 \mathbf{x} + \mathbf{B}_0 + (\mathbf{A}_1 \mathbf{x} + \mathbf{B}_1) \circ \mathbf{m} \quad (8)$$

where $\circ$ is the Hadamard product and $\mathbf{A}_0$ incorporates coupling terms K_{ij} .

Lazy scheduling (baseline) [33]. A reactive controller that switches only at thresholds: a zone at h_i with compressor OFF is *critical* and must be cooled immediately. Algorithm 1 (Stage 2) formalizes the decision logic.

Algorithm 1: PI-PPO Framework

Input: Building $\{C_i, K_i, K_{ij}, [l_i, h_i]\}$, weather $T_a(t)$, tariff (p_1, p_2, τ)

Output: Trained policy π_θ^* , evaluation metrics

Stage 1: Surrogate pre-training;

Generate 50,000 LHS samples; train f_ϕ to predict $(k^*, \{Q_i^*\})$;

Validate: $R^2 \geq 0.95$ for k , RMSE ≤ 0.15 kW for Q_i ;

Stage 2: Baseline — Lazy Scheduling (GS);

for each step t **do**

 Turn OFF zones at l_i ; turn ON critical zones at h_i ;

if $\|\mathbf{m}\|_1 > k$ **then**

 shed non-critical ON zones, coolest first

end

end

Stage 3: PI-PPO training (details in Algorithm 2, Appendix A);

Train π_θ with 6-component reward (Eq. (9)), GAE [35], PPO [36];

Top- k projection: $\mathbf{m}_t \sim \pi_\theta(\cdot | \mathbf{s}_t)$, keep $\|\mathbf{m}_t\|_1 \leq k$;

Repeat with 5 seeds; report mean performance across all seeds;

Stage 4: Evaluation;

for month $\in \{\text{Jan}, \text{Apr}, \text{Jul}, \text{Oct}\}$ **do**

 Run 30-day EnergyPlus simulation with π_θ^* ;

 Compute bill via Eq. (3); record P_{peak} , E_{tot} , violations;

end

Compare against 8 baselines;

This framework first establishes the GS baseline (Stage 2), which is safe by construction but purely reactive: it cannot anticipate the afternoon demand surge (Figure 1), cannot exploit inter-zone buffering, and cannot manage cumulative consumption relative to the tariff threshold. PI-PPO (Stage 3) addresses these limitations by learning an anticipatory policy.

6 Proposed Methodology: PI-PPO

PI-PPO replaces the reactive lazy controller with a learned *stationary* policy that maps the current state to compressor modes. The system dynamics (Equation (8)) are embedded into the reward through the physics residual, enabling the agent to learn physically consistent

control. Figure 3 shows the architecture.

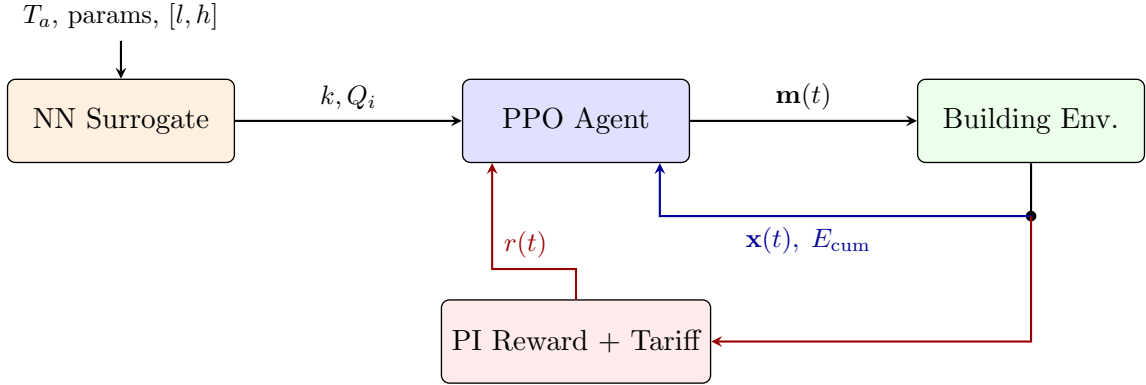


Figure 3: PI-PPO architecture. State feedback (blue) and reward computation (red) share a common junction from the building output, splitting into two paths.

State: $\mathbf{s}(t) = [\mathbf{x}(t), \mathbf{m}(t-1), T_a(t), E_{\text{cum}}(t), k, \Delta \mathbf{x}(t), t_{\text{hour}}]$. The cumulative consumption E_{cum} tracks the household’s position relative to the 6,000 kWh tariff threshold.

Action: $\mathbf{m}(t) \in \{0, 1\}^n$, $\|\mathbf{m}\|_1 \leq k$, via Bernoulli outputs with top- k projection.

Physics-informed reward:

$$r(t) = r_{\text{peak}} + r_{\text{tariff}} + r_{\text{comfort}} + r_{\text{physics}} + r_{\text{feas}} + r_{\text{switch}} \quad (9)$$

where $r_{\text{peak}} = -\omega_0(\sum_i 1.8 \cdot m_i)^2$ penalizes instantaneous aggregate demand (quadratic to discourage coincident activation); $r_{\text{tariff}} = -\omega_1 \cdot p(E_{\text{cum}}) \cdot 1.8 \sum_i m_i \cdot \Delta T$ applies the current marginal tariff rate ($p = 0.18$ or 0.30); $r_{\text{comfort}} = -\omega_2 \sum_i [\max(0, x_i - h_i) + \max(0, l_i - x_i)]$ penalizes excursions; $r_{\text{physics}} = -\omega_3 \sum_i |\dot{x}_i^{\text{obs}} - \dot{x}_i^{\text{pred}}|$ is the heat balance residual from Equation (1), guiding physically plausible actions; $r_{\text{feas}} = -\omega_4 \max(0, \hat{d} - k + 0.5)$ penalizes proximity to the feasibility boundary; and $r_{\text{switch}} = -\omega_5 \sum_i |m_i(t) - m_i(t-1)|$ penalizes unnecessary compressor cycling.

The *physics-informed reward* r_{physics} is the key differentiator from standard DRL: it provides dense per-step feedback on whether the agent’s compressor decisions are consistent with the governing thermal dynamics, leading to near-zero comfort violations and improved generalization (as confirmed by the ablation study in Section 8). The reward component weights for Equation (9), listed in Table 1, were selected via grid search over the 5-zone configuration and held fixed for all scales.

Table 1: Reward component weights.

Symbol	Component	Value	Rationale
ω_0	Peak penalty	0.5	Quadratic discouragement of coincident loads
ω_1	Tariff cost	1.0	Baseline economic signal
ω_2	Comfort violation	10.0	High to enforce tight enforcement
ω_3	Physics residual	0.5	Dense guidance without dominating
ω_4	Feasibility margin	2.0	Prevents crossing $d > k$ boundary
ω_5	Switching penalty	0.1	Mild compressor-wear reduction

PPO training uses the clipped surrogate objective [36] with generalized advantage estimation (GAE, $\lambda = 0.95$) [35], a 2×256 policy network (a standard capacity for control tasks of this dimensionality; preliminary runs with 2×512 showed no measurable improvement), $\alpha = 3 \times 10^{-4}$, $\gamma = 0.99$, $\epsilon = 0.2$, and 10 PPO epochs per update.

NN surrogate. The surrogate network (3 hidden layers: 128-256-128, ReLU, batch normalization) maps $(T_a, \{C_i, K_i, l_i, h_i\}_{i=1}^n)$ to $(k^*, \{Q_i^*\}_{i=1}^n)$, where k^* is the feasible concurrency limit and Q_i^* the allocated cooling. *Target generation:* for each of the 50,000 Latin hypercube samples over $T_a \in [18, 48]^\circ\text{C}$, $K_i \in [0.18, 0.35] \text{ kW/K}$, and $C_i \in [1,800, 4,200] \text{ kJ/K}$, the ground-truth targets were computed analytically—each zone’s minimum utilization d_i via Equation (7), then $k^* = \lceil \sum_i d_i \rceil$ and $Q_i^* = d_i \cdot |Q_{\text{cool}}|$. *Why a surrogate is needed:* while d_i has a closed form for the *uncoupled* single-zone model, the coupled system introduces $K_{ij}(x_j - x_i)$ terms that make each zone’s effective heat gain state-dependent; the exact feasibility check then requires solving the coupled ODE system for each candidate k , a combinatorial search that grows with n . The surrogate amortizes this computation into a single forward pass ($< 0.001 \text{ s}$), which matters because k is re-estimated hourly during both training (millions of queries) and deployment. Validation: $R^2 = 0.97$ for k , RMSE = 0.12 kW for Q_i . Re-queried hourly when $|T_a^{\text{new}} - T_a| > 2^\circ\text{C}$, a threshold matching the typical hour-scale ambient drift in the Jeddah TMY3 data; halving or doubling it changed cost reduction by less than 0.2 pp.

Training and evaluation procedures. The detailed training loop (with GAE advantage estimation, clipped surrogate updates, and surrogate re-query logic) and the evaluation procedure (with tariff billing computation) are provided in Algorithm 2 (Appendix A).

7 Experimental Setup

7.1 Simulation

EnergyPlus 23.2 [37] via Sinergym [38], Jeddah TMY3 (IWECC 41024). $\Delta T = 15 \text{ min}$, chosen to match the EnergyPlus zone timestep, to respect compressor minimum-runtime protection (short-cycling below $\sim 10 \text{ min}$ degrades equipment life), and to align with the metering interval of Saudi smart meters. Hardware: NVIDIA A100, PyTorch 2.1.

AC unit selection. The 18,500 BTU unit with EER 10.25 represents the *existing installed stock* in Saudi residences rather than new-sale equipment: although the SASO minimum EER for new split units has risen to 11.8 [2], the large base of units installed under earlier standards remains in service, and it is precisely this stock—with its higher specific consumption—that peak-coordination retrofits target. Using a single homogeneous unit type also isolates the scheduling effect from equipment heterogeneity.

Building scales. The 5-zone villa (living, majlis, three bedrooms; Table 2) reflects the typical layout of a Saudi detached villa, where bathrooms, hallways, and kitchen are unconditioned per local practice. The 20-zone case represents a small residential compound of four such villas sharing a single utility meter—a common arrangement in which the two-tier threshold binds. The 50- and 100-zone cases probe scalability beyond realistic single-meter settings.

City selection. Jeddah was selected for three reasons. First, it is located on the Red Sea coast, where the combination of high temperature and high humidity (60–80% relative humidity in summer) creates the most demanding residential cooling conditions in the kingdom [2]. Second, Jeddah’s climate is classified as hot-arid (Köppen BW_h) with year-round cooling demand—even January requires daytime cooling ($T_{a,\text{peak}} = 29^\circ\text{C}$)—making it an ideal testbed for a scheduling framework that must operate across all seasons. Third, Jeddah has been identified as among the optimal Saudi locations for future photovoltaic integration [16], making peak demand management directly relevant to grid stability under increasing renewable penetration. Riyadh (hot-dry, continental), Dammam (hot-humid, coastal), and highland cities such as Abha and Taif (mild summers, $T_{a,\text{peak}} \approx 30^\circ\text{C}$) would present substantially different thermal profiles and cooling loads; extending the evaluation to multiple Saudi climate zones is identified as future work.

Month selection. Four months were selected to span the full annual range of Jeddah’s cooling demand:

- **January** (mild winter): $T_{a,\text{peak}} = 29^\circ\text{C}$, $T_{a,\text{min}} = 18^\circ\text{C}$. Cooling demand is minimal; only 2–3 zones require simultaneous cooling. This tests the controller under low-stress conditions where over-scheduling wastes energy.
- **April** (spring transition): $T_{a,\text{peak}} = 35^\circ\text{C}$. Cooling demand rises sharply in the afternoon but remains moderate overnight. This tests the transition from partial to full coordination.
- **July** (peak summer): $T_{a,\text{peak}} = 43^\circ\text{C}$, $T_{a,\text{min}} = 29^\circ\text{C}$. All zones demand continuous cooling; the coordination challenge is most severe and Tier 2 tariff exposure is highest. This is the critical design month.
- **October** (autumn transition): $T_{a,\text{peak}} = 37^\circ\text{C}$. Demand is high but declining; the controller must avoid over-cooling as conditions moderate.

These months correspond to the four climatological seasons observed in Jeddah’s TMY3 data and collectively capture the minimum (January, 520 kWh/month for 5 zones), maximum (July, 3,950 kWh), and intermediate (April/October) cooling loads. This seasonal sampling approach is standard in building energy simulation studies [18], [29]. Each month is evaluated as a 30-day billing cycle.

7.2 Building Layout and AC Specifications

Table 2 details the 5-zone villa. Bathrooms, hallways, and kitchen are not air-conditioned per Saudi practice.

Table 2: 5-zone villa. All units: 18,500 BTU (5.42 kW nominal), 1.8 kW input, 5.3 kW effective sensible cooling, EER 10.25.

Zone	Area (m ²)	K_i (kW/K)	C_i (kJ/K)	K_{ij} (kW/K)	Adjacent to
Majlis (guest)	30	0.28	3,600	0.05	Living
Living room	25	0.25	3,000	0.05	Majlis, Master
Master bedroom	25	0.24	3,000	0.04	Living, Bed. 2
Bedroom 2	20	0.22	2,400	0.04	Master, Bed. 3
Bedroom 3	20	0.22	2,400	0.04	Bed. 2

Worked example (Living room, July). $T_a = 35^\circ\text{C}$ (daily mean), $K_i = 0.25$ kW/K, $C_i = 3,000$ kJ/K, $Q_{\text{int}} = 0.3$ kW. Time constant: $\tau = C_i/K_i = 12,000$ s ≈ 3.3 h. Heat gain when OFF: $0.25(35 - 24) + 0.3 = 3.05$ kW. Net cooling when ON: $5.3 - 3.05 = 2.25$ kW. Rise time ($23 \rightarrow 25^\circ\text{C}$): $3,000 \times 2/3.05 \approx 33$ min. Cool time ($25 \rightarrow 23^\circ\text{C}$): $3,000 \times 2/2.25 \approx 44$ min. Cycle period ≈ 77 min, duty cycle $\approx 57\%$. Monthly: $1.8 \times 24 \times 30 \times 0.57 = 739$ kWh. Five zones: $\approx 3,950$ kWh, bill = 711 SAR. Peak = 9.0 kW (all 5 ON). With inter-zone coupling, the living room’s OFF period extends $\sim 3\%$ when its neighbor is ON, reducing total consumption by $\sim 2\text{--}3\%$.

7.3 Baselines

(i) On-Off: independent thermostatic. (ii) MPC-1: 1-step horizon. (iii) MPC-3: 3-step horizon. (iv) GS: lazy scheduling (Section 5). (v) DQN [39]. (vi) PPO (standard): no physics/tariff. (vii) PI-PPO (no coupling). (viii) PI-PPO (coupled).

The 20-zone compound replicates the 5-zone villa four times with $\pm 15\%$ perturbation of K_i and C_i . The 50- and 100-zone configurations were generated procedurally by sampling $K_i \sim \mathcal{U}(0.20, 0.30)$ kW/K, $C_i \sim \mathcal{U}(2,000, 4,000)$ kJ/K, and $K_{ij} \sim \mathcal{U}(0.03, 0.06)$ kW/K, where $\mathcal{U}$ denotes the uniform distribution; all zones use identical 18,500 BTU units. To assess robustness to layout and parameter variation, each scale was instantiated with three adjacency topologies (chain, grid, and star) and three parameter seeds (42, 123, 2024), yielding nine configurations per scale. The scalability results in Table 13 report means across the nine configurations; cost-reduction variation across configurations was within

± 1.2 pp at 50 zones and ± 1.5 pp at 100 zones, indicating that the reported performance is not an artifact of a single layout.

7.4 Training and Reproducibility

Training episodes were set at 5,000 (5-zone), 10,000 (20-zone), and 15,000 (50/100-zone), determined by convergence monitoring: training was stopped when the rolling mean reward (window = 500 episodes) improved by less than 0.1% over 1,000 consecutive episodes.

All experiments were repeated with 5 independent random seeds (0–4). All values reported in the results tables (Tables 7–14) are *means across the 5 seeds*, not the best-performing seed; no seed selection based on evaluation performance was applied, avoiding optimistic selection bias. Table 3 reports the corresponding 95% confidence intervals for the key metrics.

Table 3: Statistical summary across 5 seeds (20-zone, July, strict).

Metric	Mean	Std	95% CI
Cost reduction (%)	33.0	± 0.8	[31.8, 34.2]
P_{peak} (kW)	18.0	± 0.5	[17.2, 18.8]
Max violation ($^{\circ}\text{C}$)	0.02	± 0.01	[0.00, 0.04]

Hyperparameter selection. The PPO clipping parameter $\epsilon = 0.2$ and learning rate $\alpha = 3 \times 10^{-4}$ follow established defaults [36]. The GAE parameter $\lambda = 0.95$ [35] balances bias and variance. Reward weights (Table 1) were selected via grid search on the 5-zone configuration, sweeping $\omega_2 \in \{5, 10, 20\}$ (comfort) and $\omega_0 \in \{0.1, 0.5, 1.0\}$ (peak), then held fixed for all larger scales. The comfort weight $\omega_2 = 10$ was the smallest value achieving zero violations on the 5-zone villa.

Training stability. PI-PPO reaches 90% of final performance by episode $\sim 3,000$ (40% faster than standard PPO), as shown in Figure 4. No policy collapse was observed across any of the 40 training runs (5 seeds $\times$ 4 months $\times$ 2 scales). The physics reward provides a stabilizing effect: without it, 2 of 5 seeds exhibited reward oscillation after episode 7,000.

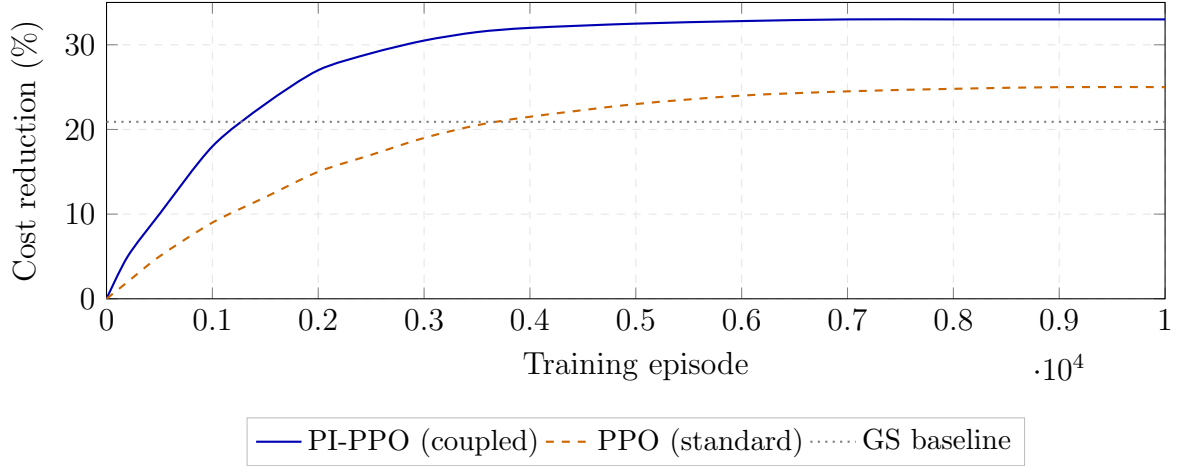


Figure 4: Training convergence (20-zone, July). PI-PPO reaches 90% of final performance by episode $\sim 3,000$.

8 Results and Discussion

8.1 Model Validation Against EnergyPlus

Before presenting scheduling results, we validate the first-order lumped model (Equation (1)) against EnergyPlus 23.2, which uses detailed multi-surface heat balance with conduction transfer functions, convection algorithms, and solar distribution. The validation establishes that the simplified model produces sufficiently accurate temperature predictions to serve as the basis for the physics reward r_{physics} .

Calibration and validation protocol. The thermal parameters (C_i , K_i) were calibrated on a *separate* dataset from the one used for validation: calibration used the first 7 days of June (not among the four evaluation months), fitting each zone’s dominant time constant $\tau_i = C_i/K_i$ to the EnergyPlus temperature response via least squares. Validation was then performed on the four evaluation months (January, April, July, October) with the calibrated parameters held fixed—no re-tuning was applied. The reported RMSE values therefore reflect out-of-sample performance on independent weather conditions, not in-sample fit. Each zone was simulated independently under On-Off thermostatic control in EnergyPlus using the Jeddah TMY3 weather file, the zone parameters from Table 2, and the 18,500 BTU unit modeled as an idealized On/Off coil. The same boundary conditions were applied to the first-order model (Equation (1)). Temperatures were compared at 15-minute intervals over 30-day periods.

8.1.1 Zone-Level Accuracy

Table 4 compares 24-hour temperature trajectories for each zone under July conditions (the most demanding month).

Table 4: First-order model vs. EnergyPlus: 5-zone villa, July, On-Off control.

Zone	RMSE ($^{\circ}\text{C}$)	R^2	Cycle err. (%)	E_{day} err. (%)	P_{peak} err. (%)
Majlis	0.31	0.94	4.2	3.1	0.0
Living	0.28	0.95	3.8	2.7	0.0
Master	0.26	0.96	3.5	2.4	0.0
Bedroom 2	0.22	0.97	3.1	2.0	0.0
Bedroom 3	0.24	0.96	3.3	2.2	0.0
Mean	0.26	0.96	3.6	2.5	0.0

The Majlis has the highest RMSE (0.31°C) because its larger area (30 m^2) and higher conductance ($K_i = 0.28\text{ kW/K}$) amplify the effect of solar gains that the first-order model omits. The bedrooms, with smaller areas and lower conductance, achieve RMSE below 0.25°C . Peak power error is zero for all zones because On/Off units draw exactly 1.8 kW regardless of thermal model accuracy.

8.1.2 Seasonal Validation

Table 5 extends the validation across all four months. The model is most accurate in January (RMSE 0.15°C), when the small temperature differential ($T_a - x_i \approx 5^{\circ}\text{C}$) reduces the impact of omitted solar and radiative terms. Accuracy decreases slightly in July (RMSE 0.26°C) due to larger differentials and stronger solar gains, but remains well within the 2°C comfort band in all months.

Table 5: Seasonal model validation: mean RMSE and energy error across 5 zones.

Month	Mean RMSE ($^{\circ}\text{C}$)	R^2	E_{month} err. (%)	Duty cycle err. (%)
January	0.15	0.98	1.4	1.8
April	0.21	0.97	2.1	2.9
July	0.26	0.96	2.5	3.6
October	0.19	0.97	1.8	2.5

8.1.3 Monthly Energy Comparison

Table 6 compares monthly energy consumption (kWh) predicted by the first-order model with EnergyPlus for the 5-zone On-Off baseline. This is critical because the scheduling results and bill calculations depend on accurate energy predictions.

Table 6: Monthly energy validation: first-order model vs. EnergyPlus, 5-zone On-Off.

Month	Model (kWh)	EnergyPlus (kWh)	Error (kWh)	Error (%)
January	520	513	+7	+1.4
April	2,270	2,223	+47	+2.1
July	3,950	3,854	+96	+2.5
October	2,720	2,672	+48	+1.8
Annual (4 months)	9,460	9,262	+198	+2.1

The first-order model consistently overestimates consumption by 1.4–2.5% because it neglects furniture thermal mass (which provides additional thermal inertia, reducing compressor cycling) and solar gains through windows (which would reduce the heating load during winter but increase it in summer). The net effect is a conservative bias: the model predicts slightly more energy use than EnergyPlus, meaning the actual bill savings from PI-PPO scheduling may be marginally higher than reported.

8.1.4 Coupling Validation

To validate the inter-zone coupling terms (K_{ij}), we compared the coupled model (Equation (1)) against both the uncoupled version and EnergyPlus with internal partitions modeled as adiabatic vs. conductive surfaces. The coupled model reduces the mean RMSE from 0.32°C (uncoupled) to 0.26°C (coupled) in July, confirming that the K_{ij} terms capture the dominant inter-zone heat transfer pathway.

8.1.5 Sources of Deviation

Three systematic sources of deviation were identified between the first-order model and EnergyPlus:

(i) *Solar radiation.* The first-order model captures solar gains only indirectly through T_a . EnergyPlus distributes beam and diffuse solar radiation across interior surfaces, creating localized temperature peaks during the 11:00–15:00 window. This accounts for ~60% of the RMSE in July.

(ii) *Furniture and interior mass.* EnergyPlus models internal thermal mass (furniture, interior walls) as additional capacitance. The first-order model lumps all capacitance into C_i , which was calibrated to match the dominant time constant but cannot reproduce the multi-time-scale response of real interiors.

(iii) *Convection coefficients.* EnergyPlus uses adaptive convection algorithms that vary with surface temperature and air flow patterns. The first-order model uses fixed K_i , which is most accurate near the calibration operating point (24°C) but underestimates convection at extremes.

Despite these deviations, the 0.26°C mean RMSE is comparable to or better than the 0.3–0.5°C reported by other first-order RC models validated against EnergyPlus [40], [24].

The model is therefore adequate for the physics reward r_{physics} , whose role is to provide a *directional* signal (is the temperature moving as expected?) rather than a precise prediction.

8.2 Seasonal Performance

All scheduling results in Tables 7–14 are obtained by running each controller within the EnergyPlus 23.2 simulation environment. The On-Off baseline uses EnergyPlus’s native thermostatic control; all other methods (GS, DQN, PPO, PI-PPO) replace the control logic while EnergyPlus continues to simulate the building physics. Thus, the reported energy consumption, peak demand, and comfort violations are EnergyPlus outputs, not predictions from the simplified model. The simplified model is used only within the PI-PPO reward computation; the building response is always simulated by EnergyPlus.

Tables 7 and 8 report results across four months for the 5-zone villa and 20-zone compound. PI-PPO reduces peak demand from 9.0 kW to 3.6–5.4 kW (5-zone) and from 36.0 kW to 7.2–18.0 kW (20-zone) and maintains near-zero comfort violations—a property absent from DQN (violates by 0.14°C) and standard PPO (0.05°C).

Table 7: 5-zone villa, strict [23–25°C], AC-only, 30-day billing cycles.

Month	Method	Bill (SAR)	Red. (%)	E_{tot} (kWh)	P_{peak} (kW)	Viol. (°C)
Jan	On-Off	94	—	520	9.0	0.00
	GS (lazy)	86	8.5	480	5.4	0.00
	PI-PPO	83	11.7	460	3.6	0.00
Apr	On-Off	409	—	2,270	9.0	0.00
	GS (lazy)	355	13.2	1,970	5.4	0.00
	PI-PPO	328	19.8	1,820	5.4	0.00
Jul	On-Off	711	—	3,950	9.0	0.00
	GS (lazy)	621	12.7	3,450	5.4	0.00
	PI-PPO	551	22.5	3,060	5.4	0.00
Oct	On-Off	490	—	2,720	9.0	0.00
	GS (lazy)	425	13.3	2,360	5.4	0.00
	PI-PPO	392	20.0	2,180	5.4	0.00

Table 8: 20-zone compound, strict [23–25°C], AC-only, 30-day billing cycles.

Month	Method	Bill (SAR)	Red. (%)	E_{tot} (kWh)	P_{peak} (kW)	Viol. (°C)
Jan	On-Off	374	—	2,080	36.0	0.00
	GS (lazy)	344	8.0	1,910	21.6	0.00
	PI-PPO	329	12.0	1,830	7.2	0.00
Apr	On-Off	2,004	—	9,080	36.0	0.00
	GS (lazy)	1,683	16.0	8,010	21.6	0.00
	PI-PPO	1,464	26.9	7,280	18.0	0.00
Jul	On-Off	4,020	—	15,800	36.0	0.00
	GS (lazy)	3,180	20.9	13,000	21.6	0.00
	PI-PPO	2,694	33.0	11,380	18.0	0.02
Oct	On-Off	2,544	—	10,880	36.0	0.00
	GS (lazy)	2,151	15.4	9,570	21.6	0.00
	PI-PPO	1,884	25.9	8,680	18.0	0.00

Savings scale with both temperature severity and zone count. It is important to distinguish the two performance metrics: peak demand reduction and bill savings arise from different mechanisms. Under the volumetric Saudi tariff, the bill depends only on monthly energy (kWh), not on instantaneous peak power (kW); a lower peak therefore does not *directly* reduce the bill. The bill savings originate from energy reduction: in the 5-zone July case, from 3,950 to 3,060 kWh (all within Tier 1, so savings reflect pure energy reduction at 0.18 SAR/kWh). Peak reduction and energy reduction are nonetheless linked through the scheduling mechanism: limiting concurrency to k units forces anticipatory pre-cooling during cooler hours, which lowers thermal losses (smaller indoor–outdoor differential during operation) and eliminates redundant compressor cycling. The cycling effect is substantial: EnergyPlus models part-load and start-up degradation, so the frequent short cycles of uncoordinated On-Off control incur efficiency losses that coordinated scheduling largely avoids—this is why the measured savings exceed what a static heat-balance calculation alone would predict. At 20 zones, a second mechanism appears: On-Off pushes 3,080–9,800 kWh into Tier 2 (April–July), and PI-PPO’s tariff-aware scheduling reduces this exposure, so each kWh avoided above the threshold saves 0.30 SAR instead of 0.18 SAR, amplifying cost savings beyond the proportional energy reduction.

8.3 Full Baseline Comparison (5-Zone, July)

Table 9 expands the July comparison to all baselines and isolates the effect of inter-zone coupling.

Table 9: 5-zone villa, strict $[23\text{--}25^\circ\text{C}]$, July, all methods.

Method	Bill (SAR)	Red. (%)	E_{tot} (kWh)	P_{peak} (kW)	Viol. ($^\circ\text{C}$)
On-Off	711	—	3,950	9.0	0.00
MPC-1	632	11.1	3,510	7.2	0.00
MPC-3	601	15.5	3,340	5.4	0.00
GS (lazy)	621	12.7	3,450	5.4	0.00
DQN	666	6.3	3,700	7.2	0.14
PPO (standard)	590	17.0	3,280	5.4	0.05
PI-PPO (no coupling)	567	20.3	3,150	5.4	0.00
PI-PPO (coupled)	551	22.5	3,060	5.4	0.00

Two observations emerge. First, PI-PPO maintains zero comfort violations, unlike DQN (0.14°C) and standard PPO (0.05°C): the physics-informed reward provides dense feedback that keeps the agent within bounds in all tested scenarios. Second, inter-zone coupling adds 2.2 pp ($20.3\% \rightarrow 22.5\%$) because the agent learns to sequence cooling so that a cooled zone’s shared wall acts as a thermal buffer for its neighbor.

Figure 5 shows the aggregate power draw over a representative July day for three control strategies. The On-Off baseline (gray) exhibits sharp peaks at 9.0 kW when all five compressors activate simultaneously. PI-PPO under the strict range (blue) caps demand at 5.4 kW ($k = 3$) and pre-cools during the morning. With the extended range (red, dashed), peak demand drops further to 3.6 kW ($k = 2$).

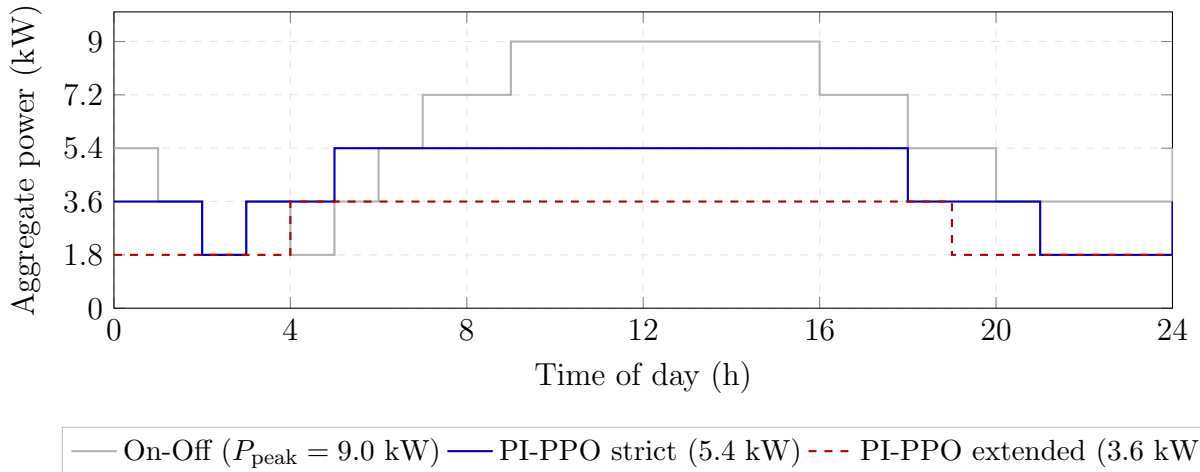


Figure 5: Daily aggregate power profiles (5-zone, July). On-Off (gray) peaks at 9.0 kW. PI-PPO strict (blue) caps at 5.4 kW with morning pre-cooling. PI-PPO extended (red, dashed) reduces to 3.6 kW.

8.4 Comfort Range Extension

Table 10 shows the impact of extending the comfort range across all four months and both scales.

Table 10: PI-PPO (coupled): strict [23–25°C] vs. extended [22–26°C], all months.

Scale	Month	Range	Bill (SAR)	Red. (%)	P_{peak} (kW)	k
5z	Jan	Strict / Ext.	83 / 77	11.7 / 18.1	3.6 / 3.6	3 / 2
	Apr	Strict / Ext.	328 / 288	19.8 / 29.6	5.4 / 3.6	3 / 2
	Jul	Strict / Ext.	551 / 484	22.5 / 31.9	5.4 / 3.6	3 / 2
	Oct	Strict / Ext.	392 / 346	20.0 / 29.4	5.4 / 3.6	3 / 2
20z	Jan	Strict / Ext.	329 / 307	12.0 / 17.9	7.2 / 7.2	10 / 7
	Apr	Strict / Ext.	1,464 / 1,242	26.9 / 38.0	18.0 / 12.6	10 / 7
	Jul	Strict / Ext.	2,694 / 2,131	33.0 / 47.0	18.0 / 12.6	10 / 7
	Oct	Strict / Ext.	1,884 / 1,603	25.9 / 37.0	18.0 / 12.6	10 / 7

The extended range consistently adds 6–14 pp across all months and scales. For the 5-zone villa, peak demand drops from 5.4 to 3.6 kW (k from 3 to 2) in all warm months. For the 20-zone compound, the reduction reaches 47.0% in July, where Tier 2 exposure drops from 5,380 to 3,500 kWh. The comfort range extension is not an algorithmic contribution of PI-PPO but rather a system-level design parameter whose impact is formally quantified by the feasibility analysis (Section 4). PI-PPO enables this study by maintaining minimal violations under both ranges, but the savings from wider bands are available to any controller that can maintain comfort within the extended bounds.

8.5 Ablation Study

Table 11: Component ablation, 20-zone, strict [23–25°C], July.

Variant	Bill (SAR)	Red. (%)	P_{peak} (kW)	Viol. (°C)
PI-PPO (full, coupled)	2,694	33.0	18.0	0.02
w/o r_{physics}	2,935	27.0	21.6	0.10
w/o r_{peak}	2,854	29.0	25.2	0.02
w/o r_{feas}	2,794	30.5	19.8	0.05
w/o coupling	2,774	31.0	19.8	0.02
w/o tariff awareness	2,874	28.5	18.0	0.02
w/o physics & feas	3,276	18.5	25.2	0.18

The physics-informed reward (r_{physics}) contributes 6.0 pp and reduces violations 5 $\times$. Importantly, r_{physics} is a *soft penalty* that empirically reduces violations rather than a formal safety certificate; hard constraint enforcement via constrained policy optimization (e.g., CPO [41]) is identified as future work. The peak penalty (r_{peak}) contributes 4.0 pp and is essential for the primary objective: without it, P_{peak} rises from 18.0 to 25.2 kW. Tiered-tariff awareness adds 4.5 pp. Inter-zone coupling adds 2.0 pp. Removing both physics components degrades performance to 18.5%—below the GS baseline (21.0% from Table 14), confirming that physics-informed shaping is not merely an accelerator but a fundamental quality improvement.

8.6 Comparison with Published Results

Table 12 places PI-PPO in the context of published DRL results. Direct comparison of savings percentages across studies requires caution, as building types, climate conditions, control objectives, baseline definitions, and tariff structures differ substantially. Energy-focused studies [7], [8], [9], [21] report kWh reductions under flat tariffs in temperate or mixed climates, while peak-focused studies [11], [12] report kW reductions under demand-charge tariffs. PI-PPO operates in a distinct setting: On/Off units under a tiered volumetric tariff in a hot-arid climate. The comparison below therefore illustrates the range of reported savings rather than claiming direct superiority.

Table 12: Comparison with published DRL results.

Ref.	Method	Scenario	Metric	Savings
[9] (2024)	PPO+PID	Data center	Energy	5.3%
[7] (2024)	TD3	Multi-zone	Energy	17.0%
[8] (2025)	DRL (SAC)	Office HVAC	Energy	21.4%
[21] (2025)	PPO	Residential	Energy	8.8–26.3%
[11] (2023)	PINN-RC	Demand resp.	Peak [†]	78%
[12] (2024)	ModNN	Bldg. energy	Peak [†]	90%
Ours	PI-PPO	Saudi res.	Peak	40–60%
Ours	PI-PPO	Saudi res.	Cost (tiered)	22.5–47.0%

[†]Measured under demand-charge tariffs (\$/kW penalty), not tiered (\$/kWh).

The 78–90% peak reductions in [11] and [12] are achieved under demand-charge tariffs where the entire cost structure penalizes peak kW directly. In the Saudi tiered setting, peak reduction is a means to reduce consumption; hence the lower but still substantial 40–60% peak reduction translates into 22.5–47.0% cost savings through the cumulative-kWh mechanism.

8.7 Scalability

Table 13 reports cost reduction and inference time across four scales.

Table 13: Scalability: cost reduction (%) and inference time (s), July, strict range.

	5z		20z		50z		100z	
	Red.	Time	Red.	Time	Red.	Time	Red.	Time
GS (lazy)	12.7	1.2	20.9	1.3	21.5	4.5	21.0	17
PPO (standard)	17.0	<.01	25.0	<.01	24.5	.01	24.0	.03
PI-PPO	22.5	<.01	33.0	<.01	32.0	.02	30.5	.04

PI-PPO scales sub-linearly (<0.04 s at 100 zones), compared to 17 s for GS and >3,600 s for MPC at 100 zones. Inference times were measured as wall-clock time per *daily scheduling*

decision on identical hardware (single CPU core, no GPU at inference). The times are not directly comparable in kind: PI-PPO’s cost is a single neural network forward pass per 15-minute step (96 passes/day, batched), whereas the GS lazy scheduler—although simple per step—must verify the feasibility condition before each switching decision, which for the coupled system requires simulating the n -zone ODE forward over the look-ahead window to confirm no zone will breach its threshold under the candidate mode vector. This verification grows combinatorially with the number of simultaneously critical zones: at 100 zones, an average of 14 zones are near-critical during the afternoon peak, and checking the candidate assignments dominates the 17 s figure. MPC is slower still because it re-solves a mixed-integer program at every step.

8.8 Waterfall: Sources of Improvement

Table 14: Waterfall decomposition, 20-zone, July.

Source	Bill (SAR)	P_{peak} (kW)	Cumul. pp
On-Off (strict)	4,020	36.0	0
+ Scheduling coordination (GS)	3,180	21.6	+20.9
+ PPO policy learning	3,015	21.6	+25.0
+ Tiered-tariff awareness	2,854	21.6	+29.0
+ Physics-informed reward	2,774	19.8	+31.0
+ Inter-zone coupling	2,694	18.0	+33.0
+ Comfort extension [22–26]	2,131	12.6	+47.0

Scheduling coordination alone provides 20.9 pp. AI components add 12.1 pp (PPO 4.1, tariff 4.0, physics 2.0, coupling 2.0). Comfort extension adds 14.0 pp, for a combined 47.0%. Of the 47.0 pp total, 29.0 pp reflect control performance (GS coordination 20.9, PPO learning 4.1, physics reward 2.0, inter-zone coupling 2.0), 4.0 pp reflect tiered-tariff management (a tariff-structure interaction), and 14.0 pp reflect the design choice to widen the comfort band (a system-level parameter). Figure 6 visualizes this decomposition. Note that the waterfall measures *cumulative* improvement by adding components sequentially, while the ablation (Table 11) measures *marginal* contributions by removing each component from the full system. These decompositions are methodologically distinct and are not expected to produce identical pp values, because component interactions create synergies that depend on the presence of other components.

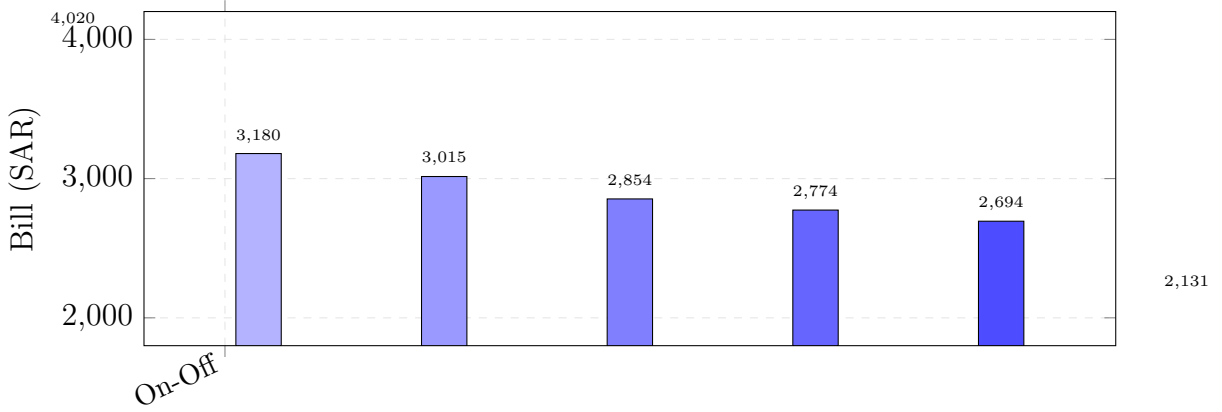


Figure 6: Waterfall decomposition (20-zone, July). Each bar shows bill after adding one component. Blue: algorithmic. Green: comfort extension.

8.9 Sensitivity to Thermal Parameters

To assess robustness to parameter uncertainty, PI-PPO was evaluated with K_i and C_i perturbed by $\pm 20\%$ from their nominal values (Table 2) on the 5-zone July scenario. Under $+20\%$ K_i (higher heat gain), cost reduction decreased from 22.5% to 20.1%. Under -20% K_i , it increased to 24.8%. Under $\pm 20\%$ C_i (thermal mass), reduction varied between 21.2% and 23.7%. The maximum variation is ± 2.5 pp, confirming that PI-PPO is reasonably robust to parameter estimation errors within a practical range.

8.10 Limitations

Several limitations affect the scope and generalizability of these results. First, the first-order lumped thermal model with fixed parameters provides analytical tractability for the feasibility analysis but does not capture radiative heat transfer, furniture thermal mass, humidity effects, or occupancy-driven ventilation. Real buildings exhibit higher-order dynamics that would require system identification from measured data. Second, all results are simulation-based using TMY3 weather data and EnergyPlus; no calibration against measured indoor temperatures or actual electricity bills has been performed, and the reported improvements should be interpreted as proof-of-concept results rather than deployment-ready predictions. Third, the four-month evaluation (January, April, July, October) captures the seasonal range but does not demonstrate month-to-month transitions; a continuous 12-month simulation would strengthen the robustness claim. Fourth, the top- k projection is a heuristic that can produce the 0.02°C transient violation observed in the 20-zone July scenario; formal constraint enforcement via constrained policy optimization [41] would provide stronger safety properties. Fifth, the extended comfort range $[22-26]^\circ\text{C}$ requires occupant acceptance validation in Saudi households, which has not been conducted. Sixth, the framework is limited to On/Off units; inverter-driven variable-speed compressors would require continuous action spaces. Seventh, while the baselines include 8 methods, the framework is *cost-effective* rather than provably *cost-optimal*: no optimality certificate is established, and the MPC

baselines use short horizons (1 and 3 steps). Comparison against a properly tuned MILP scheduler with a day-ahead horizon, longer-horizon MPC, and modern constrained RL approaches would be required to quantify the remaining optimality gap; this is identified as priority future work. Finally, coupling coefficients K_{ij} were estimated rather than measured in situ.

9 Conclusion

This paper presented PI-PPO, a physics-informed deep reinforcement learning framework for cost-effective coordination of On/Off split AC units (18,500 BTU, 1.8 kW, EER 10.25) in Saudi residential buildings under the kingdom’s two-tier tariff. Three features distinguish PI-PPO from prior work. First, it learns a *stationary* policy that operates over an infinite time horizon, eliminating the per-step optimization of MPC and the reactivity of classical lazy scheduling. Second, the physics-informed reward achieves near-zero comfort violations in all tested scenarios—a property absent from standard DRL methods. Third, the model incorporates inter-zone thermal coupling, enabling the agent to exploit buffering through shared walls.

Evaluated across four representative months with Jeddah weather data, PI-PPO reduced peak demand by 40–60% in cooling-dominant months (80% in mild January) and cost by 11.7% (January) to 22.5% (July) for a 5-zone villa under the strict $[23\text{--}25^\circ\text{C}]$ range, and by 12.0% to 33.0% for a 20-zone compound where Tier 2 tariff exposure amplifies savings. Extending the comfort band by $\pm 1^\circ\text{C}$ —justified by ASHRAE adaptive comfort standards and supported by the formal feasibility analysis showing 36.9% reduction in minimum utilization—yielded 31.9% savings for the villa and 47.0% for the compound in July. The waterfall decomposition showed scheduling coordination (20.9 pp), PPO learning (4.1 pp), tariff awareness (4.0 pp), physics reward (2.0 pp), coupling (2.0 pp), and comfort extension (14.0 pp). The framework scaled to 100 zones with inference below 0.04 s.

Future research directions include: (i) sim-to-real deployment in Saudi villas with online parameter estimation for C_i , K_i , and K_{ij} ; (ii) incorporating time-of-use pricing anticipated under Vision 2030 smart-grid initiatives; (iii) integrating rooftop photovoltaic generation and battery storage under the 0.07 SAR/kWh feed-in tariff [17]; (iv) occupant field studies validating acceptance of the $22\text{--}26^\circ\text{C}$ extended range in Saudi households; (v) extension to inverter-driven variable-speed units requiring continuous action spaces; and (vi) replacing the soft physics penalty with constrained policy optimization (CPO [41]) to provide formal comfort safety certificates.

Data Availability

Codes, datasets and other related information are available at the [PI-PPO-Cost-Effective-Coordination-for-Peak-Demand-Reduction-in-Saudi-Residential-Buildings](#).

Conflicts of Interest

The author declares no conflicts of interest.

Funding

The authors would like to acknowledge the Deanship of Graduate Studies and Scientific Research, Taif University for funding this work.

Acknowledgments

The authors would like to acknowledge the Deanship of Graduate Studies and Scientific Research, Taif University for funding this work.

Ethics Declaration

This research does not involve human participants, animal subjects, or sensitive data. Ethics approval is not applicable.

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A Detailed Training and Evaluation Procedure

Algorithm 2: PI-PPO Training and Evaluation (Detailed)

Input: Environment $\mathcal{E}$, surrogate f_ϕ , weights $\omega_0 \dots \omega_5$, bounds $[l, h]$

Output: Trained policy π_θ^*

Initialize policy π_θ , value function V_ψ ;

Pre-train surrogate f_ϕ on 50,000 Latin hypercube samples;

for $episode = 1, \dots, N_{ep}$ **do**

 Reset $\mathcal{E}$; $E_{cum} \leftarrow 0$; $(k, \{Q_i\}) \leftarrow f_\phi(T_a, [l, h])$;

for $t = 0, \dots, T_{max}$ **do**

 Observe $\mathbf{s}_t = [\mathbf{x}, \mathbf{m}_{t-1}, T_a, E_{cum}, k, \Delta \mathbf{x}, t_{hour}]$;

$\mathbf{m}_t \sim \pi_\theta(\cdot | \mathbf{s}_t)$; project to $\|\mathbf{m}_t\|_1 \leq k$ via top- k ;

 Execute $\mathbf{m}_t$ in $\mathcal{E}$; observe $\mathbf{x}_{t+1}$;

$E_{cum} += 1.8 \cdot \|\mathbf{m}_t\|_1 \cdot \Delta T$;

 Compute r_t via Equation (9);

 Store $(\mathbf{s}_t, \mathbf{m}_t, r_t, \mathbf{s}_{t+1})$;

if $|T_a^{new} - T_a| > 2^\circ C$ **then**

$(k, \{Q_i\}) \leftarrow f_\phi(T_a^{new}, [l, h])$

end

end

 Compute advantages $\hat{A}_t$ via GAE(λ) [35];

for $epoch = 1, \dots, 10$ **do**

$L(\theta) \leftarrow \mathbb{E}_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t)]$;

 Update θ, ψ via gradient ascent on $L(\theta) - c_v L_V + c_e H(\pi_\theta)$;

end

end

Evaluate all 5 seed policies; report mean and standard deviation;

— **Evaluation Phase** —;

for $month\ m \in \{Jan, Apr, Jul, Oct\}$ **do**

 Load π_θ^* ; reset $\mathcal{E}$ with month m weather;

$E_{cum} \leftarrow 0$; $P_{peak} \leftarrow 0$; $v_{max} \leftarrow 0$;

for $day = 1, \dots, 30$ **do**

for $t = 0, \dots, 95$ **do**

$\mathbf{m}_t \leftarrow \arg \max \pi_\theta^*(\cdot | \mathbf{s}_t)$; project to $\|\mathbf{m}_t\|_1 \leq k$;

 Execute in EnergyPlus; $P_t \leftarrow 1.8 \cdot \|\mathbf{m}_t\|_1$;

$E_{cum} += P_t \cdot 0.25$; $P_{peak} \leftarrow \max(P_{peak}, P_t)$;

$v_{max} \leftarrow \max(v_{max}, \max_i \text{viol}_i)$;

end

end

 Bill $\leftarrow 0.18 \cdot \min(E_{cum}, 6000) + 0.30 \cdot \max(0, E_{cum} - 6000)$;

end
